

Ensemble Learning Approach for Flow-based Intrusion Detection System

Skhumbuzo Zwane
Department of Computer Science
University of Zululand
KwaDlangezwa, South Africa
201144122@unizulu.ac.za

Paul Tarwireyi
Department of Computer Science
University of Zululand
KwaDlangezwa, South Africa
tarwireyiP@unizulu.ac.za

Matthew Adigun
Department of Computer Science
University of Zululand
KwaDlangezwa, South Africa
adigunM@unizulu.ac.za

Abstract— Network security remains a critical issue due to ongoing advancements in Information and Communication Technologies (ICT) and the concomitant rise in the number of security threats. Intrusion detection systems have emerged as an essential countermeasure to preserve network security. However, over the years, the challenge has been finding a detection mechanism that is accurate enough and has low false alarm rates. It has become increasingly difficult to attain high levels of accuracy with conventional anomaly detection systems due to the dynamic nature of network traffic patterns. In recent years machine learning based detection techniques have emerged as a viable IDS solution in comparison to pattern-based detection approaches. Some machine learning solutions that are reported to have high detection accuracy in literature are ineffective in practical situations due to the unreal nature of datasets used for evaluation. Moreover, in most cases these solutions use single classifiers which are known to be outperformed by ensemblers. This paper proposes a flow based intrusion detection method that utilizes ensemble classification machine learning techniques to analyze network flow data. A performance evaluation of the ensemble of decision tree, probabilistic and non-probabilistic classification methods is conducted. The ensemble methods; adaptive boosting, bootstrap aggregation, random forests, and majority voting were analyzed using the CIDDS-001 flow based IDS evaluation datasets. The experimental results indicate that the ensemble of decision tree based classification methods perform better when compared with the ensemble of probabilistic and non-probabilistic based classification methods. Additionally, the non-probabilistic based ensemble methods take longer to train and to classify new instances.

Keywords— CIDDS-001 dataset, Flow based IDS, Network Security, Ensemble learning, Decision Tree

I. INTRODUCTION

The need for network security mechanisms has increased over the years due to the increasing usage of information and communication technologies (ICT). Many organizations including government, businesses, and medical rely on the ICT systems. The massive use of ICT renders cyber security a major concern. Every day new threats and vulnerabilities are created or produced to attack communication systems. To address this issue, in addition to traditional security methods like encryption and firewalls, effective and efficient Intrusion Detection Systems (IDSs) are required. Intrusion detection systems are used to detect attacks and anomalies in networks. They are

capable of identifying actions that violate network specified usages and attacks threatening network lifetime. Traditional IDSs utilising deep packet inspection have Quality of Service (QoS) and computation limitations, which are introduced during traffic collection, monitoring, and traffic analysis. Most of the issues arise because deep packet inspection techniques need to analyse the entire packet data/payload, which require more processing time and computational resources [1]. While other IDS use packet-based analysis, this paper explores the viability of flows as a detection measure.

Flow-based traffic fingerprinting has been applied with greater success in other areas such as large scale OS and application detection [2], [3]. A Flow-based IDS mainly analyse network flow records to detect intrusions. Flow records contain aggregated information of packet headers, this reduce the amount of data processed by the IDS. Flow-based IDS are also applicable where the processing of complete traffic is computationally difficult. In the reviewed literature, it is perceived that flow-based inspection has fewer privacy concerns than packet-based inspection because user information or payload is protected from any intermediate scans. Moreover, for applications that use end-to-end encryption it becomes difficult for packet-based IDS to analyse the encrypted data. Hence flow-based IDS becomes more appropriate if the data is encrypted, and it introduces the opportunity for constructing intelligent intrusion detection systems.

Data Mining (DM) and Machine Learning (ML) methods have been widely adopted recently for constructing appealing and intelligent intrusion detection systems [4] [5] [6]. ML classification algorithms commonly used for intrusion detection are applied in the form of; Single classifiers, hybrid classifiers, and Ensemble classifiers [7]. Single classifiers are utilised when a single ML algorithm is used to develop an intrusion detection system. Hybrid classifiers offers a combination of more than one machine learning algorithm. Ensemble classifiers use multiple weak learners, such as, classifiers performing slightly better than a random classifier. Hybrid classifiers are preferred over single classifiers, this is because one algorithm can be used for pre-processing the samples in training set, eliminating non-representative training samples, then the results can be given to the second algorithm for pattern recognition to design a classifier, this method can vastly improve intrusion detection performance. However, more recent studies argue that ensemble

classifiers enhance accuracy, and they perform well when data is scarce [8].

This paper proposes a Flow based IDS that utilises flow data for anomaly detection. The method adopts an ensemble learning technique to analyse the flow data and classify instances. The proposed model is envisaged to be applicable in both military tactical and ad-hoc networks due to its light resource usage. The rest of the paper is organised as follows; Section II provides the background and related works. Section III, describes the proposed flow based ID method, the evaluated ensemble methods and the datasets used are also described in this section. Section IV presents the experiments, and the results and discussion are presented in Section V. Finally, the conclusions and future works are presented in section VI.

II. BACKGROUND AND RELATED WORK

A. Flow-based Intrusion Detection

Flow-based intrusion detection use network flow data to detect intrusions. [9] Presented an overview of how the accuracy and performance of packet based and flow based Network Intrusion Detection System [NIDSs] are affected by threats and attacks within high speed network environment. The authors argued that packet based NIDSs have higher resource consumption, due to the fact that they process every packet's payload. They also reported that flow based NIDS are more efficient since they have less amount of data to process. In [1] a set of flow based features for IDS that can be utilized for mobile network traffic, which is privacy aware and QoS-robust are presented.

The authors adopted the Principal Component Analysis (PCA) method for feature reduction, to reduce feature processing time. Their experiments were conducted using NSL_KDD database with 12 features extracted based on the connection flows. The results obtained from the experiment indicated F-measure of about 84 %, and experimental time of 12.9 seconds. In the mentioned study, features are reduced before presenting them to the IDS for analysis. This helps reduce the amount of data to be processed, hence yielding a QoS-robust IDS.

B. Ensemble Learning approaches

The general steps followed by all machine learning approaches include; identifying or building learning datasets, feature extraction, and classification. In machine learning, selecting the right dataset is important since the ML model can only identify anomalies based on what it has been trained with. This means that the more organic and properly prepared the dataset is, the more accurate the model will be.

In [10] three arrangements of examinations were conducted using the KDD-cup 99 datasets. The authors applied normalization to reduce the dimensionality of the datasets. Secondly, they performed feature selection using Information Gain. The third analysis was performed using an ensemble approach, bootstrapping. The authors reported that their results indicate that the average performance of ensemble approach was better than other classifiers [10]. In [7] a hybrid classification method is proposed. The authors designed a

classifier ensemble using Radial Basis Function (RBF) and Support Vector Machine (SVM), where RBF and SVM are base classifiers. The authors reported that the hybrid RBF-SVM demonstrated higher percentage of classification accuracy than the based classifiers, and the testing time was enhanced due to dimension reduction applied to the datasets.

In the study [11] an intrusion detection technique based on ensemble method of machine learning is proposed. The authors used Bagging method of ensemble with REPTree as the base class. The NSL_KDD dataset was utilized, and relevant features were selected to improve the classification accuracy. Their method was evaluated using test dataset and cross validation of 10-fold. The authors argued that their proposed method exhibited highest classification accuracy with low false positives compared with other machine learning methods.

C. Problem Formulation

In the mentioned studies different authors proposed different schemes of intrusion detection. Although flow based intrusion methods have proved to be more effective than packet based intrusion detection methods. More research on the flow based approach is still required. For example, due to the amount and volume of the collected flow data, processing and analyzing this data in real-time for intrusion detection becomes a challenge in terms of storage and processing power. In contrast, more accurate and efficient intrusion detection methods remains a requirement to ensure safe and reliable communications.

This study defines a flow based intrusion detection method that utilizes the ensemble of machine learning techniques to improve detection accuracy. In addition, in the mentioned studies, the experiments were conducted with a special focus on packet based intrusion detection. This study evaluates and analyze the performance of ensemble learning methods using flow based intrusion detection system datasets. While, most of the evaluation in literature employed the KDD cup or the derivative NSL_KDD datasets, they are considered inadequate for NIDS evaluations. In [12] the authors reported that even the data set archival authority itself strongly discouraged the use of the KDD99 dataset in 2007, while emphasizing that journals and conferences reject studies with results solely based on the KDD99 data set. In [13] they also regarded the KDD99 as not a proper representation of modern network traffic, making the dataset ineffective for the evaluation of modern NIDS methods.

At the current time, adequate alternative datasets have been proposed by researchers [13]–[15]. In this paper the CIDDS-001 flow based network IDS evaluation datasets is utilized. The CIDDS-001 is one of the state of art dataset for flow-based IDS evaluation [16]. The dataset contains unidirectional netFlow data generated from a small business environment emulated using OpenStack. These dataset represent modern network traffic flow behaviours, which help to ensure realistic results when used in flow-based IDS evaluation.

The performance evaluation approach used in this study focuses on the evaluation and comparison of the ensemble of; decision tree based, probabilistic based and non-probabilistic machine learning methods. This paper mainly focuses on evaluating the suitability of ensemble machine learning

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methods as a means of detecting network intrusions for flow based IDS.

III. METHODOLOGY

This section presents the proposed flow based IDS architecture. The different components of the system are explained respectively. The section also presents the proposed ensemble learning methods evaluated for intrusion detection tasks. Finally, the dataset used for the performance evaluation of the methods is discussed.

A. Flow based IDS architecture (FIDS)

This study proposes a flow based IDS based on ensemble machine learning method, this is because flow-based IDS have a number of advantages over traditional packet based IDS. The proposed FIDS model contains three main components, namely; Data acquisition and feature extraction, Data preprocessing, and decision engine component (Classifier). These components are discussed in this subsection.

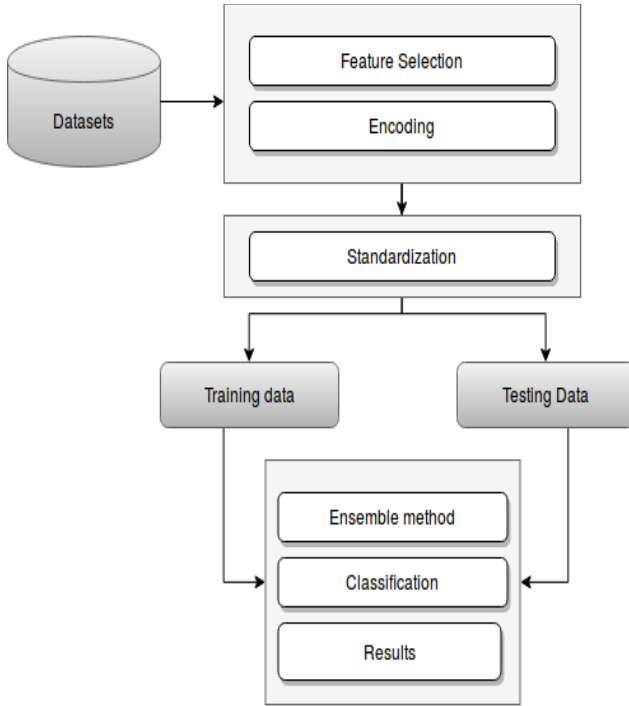


Fig. 1: Proposed Flow based IDS model

1) Data acquisition and Feature extraction

The proposed FIDS make use of network flow data (records) for intrusion detection. The task of acquiring flow records from networks have been simplified recently [12]. Major vendors now offer built-in flow collection and export support in their hardware. Examples of these flow collection and export protocols include Cisco's netflow, IPFIX and sflow. We propose employing such methods for data collection. Each network device is embedded with a flow-sampling agent. Each agent samples the flow statistics based on a defined criteria and threshold, then export the generated data to a centralized collector. The central collector collects network flow records from all the network devices in the network in real-time. The flow collector then conducts feature extraction, which include

the task of picking or selecting the optimal features that will be used by the model to classify the records successfully. The Collector then packages and export the data for storage and pass it on to the data pre-processing module.

2) Data Pre-processing

Data pre-processing is the process of converting flow records into a specific format that is acceptable to the detection algorithm used. This phase includes data cleaning, fixing missing values, data encoding, and normalization. In this component, all the features of each flow record from the data collector is encoded and scaled to the range [0:1]. This allows the Machine learning algorithm to obtain consistent detection accuracy while using less processing power. This method is ideal in power constrained environments.

3) Decision Engine

To use machine learning for intrusion detection, a machine learning model is required. In the proposed intrusion detection, a machine learning classifier is build using the preprocessed data. The decision engine phase uses the machine learning model constructed to classify new instances as malicious or normal. If an instance is classified as malicious then an alert is generated, else the instance is dropped. In addition, the decision engine logs all the instances and decisions in a log file. Such log files can be exported to a Log Management and Analysis tool, to further analyze and visualize generated alerts. This can help improve incident analysis, attack containment, and recovery.

B. Ensemble Learning techniques

Intrusion detection systems are widely adopted as means of protecting and reducing damage to information systems. One of the most important components of an IDS is the decision engine. The decision engine is tasked with analyzing the data and providing an alert when malicious activities are detected. Machine learning has been used extensively for IDS decision engine. Methods like neural networks, statistical models, rule learning, and ensemble methods are methods usually utilized for the decision engine [11].

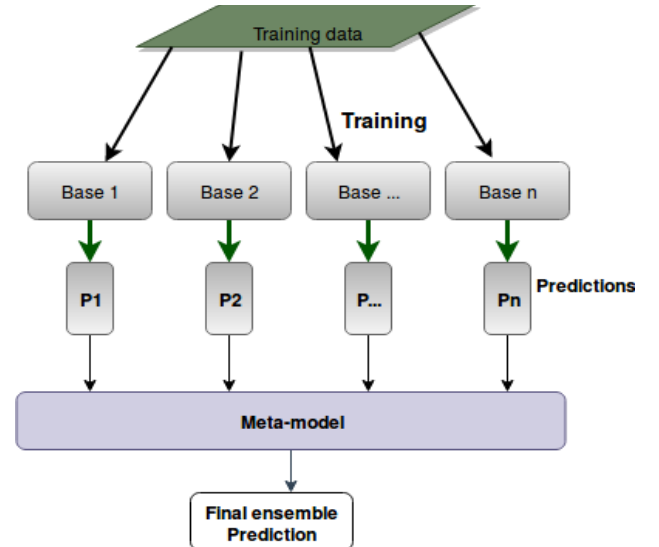


Fig.2. Ensemble classification approach

In this work, an IDS based on ensemble method is proposed, as shown in Figure 2. The philosophy behind the ensemble approach is due to the evidence that a set of classifiers perform better classification than individual classifiers (Galiano et al, 2012). In addition, ensemble machine learning methods have distinguished themselves as outstanding detector of malicious and anomalous events in intrusion detection for computer systems and networks [19]. The IDS utilizes an ensemble method that uses multiple weak learners as based methods to analyze the flow data extracted from the network.

In this paper, three representative learning methods widely used for classification problems are used as the base machine learning methods for the ensemble methods. The base methods utilized are; Decision Tree (DT), Support Vector Machine (SVM), and Naïve Bayes (NB).

1) Base Machine learning methods

Decision Tree (DT): A decision tree is a supervised learning algorithm. It is a binary structure used to classify an arbitrary input vector X . Each node in the tree represent a simple feature comparison against some field. The result of each comparison is either true or false, which is used to determine if we should proceed along the left or right child of the particular given node. Figure 3 shows a simple classification tree example.

In other words, the decision tree contain two types of nodes. First is the leaf node, which is determined by majority vote of training examples reaching that leaf. The second is the internal node, which branches out according to the answers can be used to classify a case by starting at the root of the tree and moving through it until a leaf is encountered. At each non-leaf decision node, the case's outcome for the test at the node is determined and attention shifts to the root of the sub-tree corresponding to its outcome [18]. In this study, the DecisionTreeClassifier class from scikit-learn is used. The entropy function is used for measuring the quality of a split.

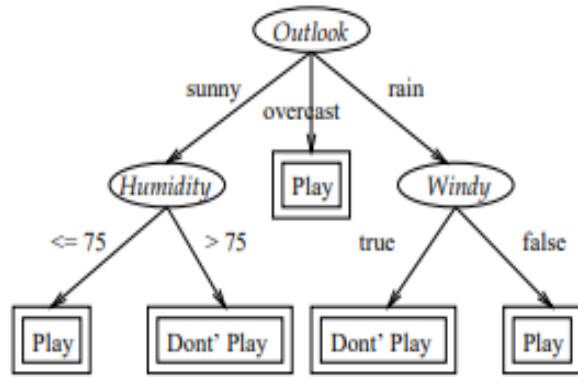


Fig 3: Classification Tree [18]

Naïve Bayes (NB): The naïve Bayes method is a set of supervised learning method based on the “Bayes Theorem” with the naïve assumption of independence between every pair of the features. More formally, suppose we want to classify a vector $X = x_1, \dots, x_n$ into m classes $C_1, \dots, C_m$. First, we need to compute the probability of each possible class given X , so that

we assign X the label of the class with the highest probability. The probability is calculated using the Bayes’ Theorem:

$$P(C_i|X) = \frac{P(X|C_i)P(C_i)}{P(X)} \quad (1)$$

Where, $P(C_i)$ is the prior probability of belonging to the class i . $P(X)$ is the normalizing constant, and $P(X|C_i)$ is the conditional probability of seeing the input vector X given we know the class is C_i .

The final prediction model is then defined by:

$$C(X) = \underset{i \in \text{classes}(t)}{\operatorname{argmax}} \frac{P(X|C_i)P(C_i)}{P(X)} \quad (2)$$

Where, $C(X)$ is the prediction returned for input X .

In other words the classifier learns from training data the conditional probability of each attribute A_i given the class label C . Classification is then done by applying Bayes rule to compute the probability of C given the particular instances of $A_1, \dots, A_n$, and then predicting the class with the highest posterior probability [19]. In this paper, we utilized the GaussianNB class provided in the scikit-learn library [17].

Support Vector Machine (SVM): SVM classifies instances by constructing a hyperplane or set of hyperplanes in a high-dimensional space.

New instances are then divided into a class based on their position in that space relative to the hyperplane [10]. The hyperplane is determined using a maximum margin hyperplane, which is the hyperplane that is the maximum distance from the training observations. Point that fall on one side of the hyperplane are classified as -1 and the other as +1. An example of the defined hyperplane is shown in Figure 4. The SVC class from scikit-learn library is used in this study.

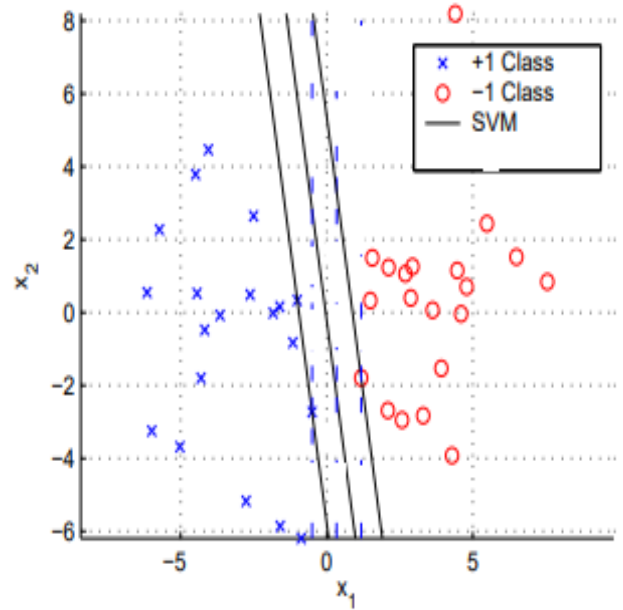


Fig 4: Example of SVM in two dimensional classification problem

2) Ensemble techniques

AdaBoost: ensemble learning method in which each new model instance focuses on training examples which were classified incorrectly in the previous models.

Bagging: ensemble learning method which aims at reducing the correlation between estimators in an ensemble via training them on random subsets of the features instead of the entire feature set.

Majority Voting: all the base classifiers are put together to vote for one class label, and the final output class label is the one that receives more than half of the votes.

Random Forest: construct a multitude of decision trees at training time and output final result based on the consideration of all the trees results.

C. Evaluation Datasets

The popularity of flow-based IDS has resulted in the proposal of different flow-based network datasets to evaluate flow-based IDS. In this paper, the Coburg Network Intrusion Detection Dataset (CIDDS-001) are utilized. The CIDDS-001 dataset is a labelled flow-based dataset containing unidirectional Netflow data [15]. The dataset consists of data extracted from an OpenStack environment with internal servers and an External server which is deployed in the internet to capture real and up-to-date traffic. The dataset consists of 14 attributes, where 1-10 are the netFlow default attributes and attribute 11 – 14 are attributes added during the labeling process. This paper utilized 11 attributes for the ensemble learning evaluation.

The CIDDS-001 dataset was captured over a period of four weeks and contains approximately 32 million flow records. It also contains 92 attacks, where 70 attacks were executed within the OpenStack environment and 22 attacks targeted to the external server. These attacks include Portscan, PingScan, DoS, and Bruteforce attacks. In this paper, the CIDDS-001-external-week1.csv datasets were used. During the process of feature extraction, out of the 14 attributes, as shown in Table I, only three were selected and used in this study. The attributes; Duration, Bytes, and Packets were selected as the most appropriate since they include numerical values that can help profile network behaviors. the selected features are further normalized using a standard scaler method provided in the scikit-learn library [19]. Each standard score x' of each sample x is calculated as:

$$x' = \frac{(x - \mu)}{\sigma} \quad (3)$$

Where, μ is the mean of the training samples, and σ is the standard deviation of the training samples.

The prepared data sets are then used to create the ensemble learning models; AdaBoost, Bagging, Random forest, and majority voting using the decision tree, probabilistic (Naïve Bayes) and non-probabilistic (Support Vector Machine) machine learning algorithms. Out of 172838 instances from the CIDDS-001-external-week1 dataset, 129628 instances were used for training each of the models and 43210 instances were used to test the models.

TABLE I. CIDDS-001 FLOW BASED DATASET ATTRIBUTES

No.	Attributes	Attribute Description
1	Src_IP_Addr	Source IP address
2	Dst_IP_Addr	Destination IP address
3	Src_port	Source port
4	Dst_port	Destination port
5	Proto	Transport protocol
6	Date_first_seen	Start time flow first seen
7	Duration	Duration of flow
8	Bytes	Number of transmitted bytes
9	Packets	Number of transmitted packets
10	Flags	OR concatenation of all TCP flags
11	Class	Class label
12	Attack_type	Type of attack
13	AttackID	Same attack class carry same attack id
14	Attack_description	Additional information about attack parameters

IV. EXPERIMENTS

A. Evaluation metrics

In machine learning evaluation, the confusion matrix is one of the most commonly used method to represent classification results [10]. A Confusion matrix is a matrix that represent true or false classification results, the different possibilities in a confusion matrix are:

- True Positive (TP): represent intrusions that are correctly classified as intrusions.
- False Positive (FP): represent normal/non-malicious instances that were misclassified.
- True Negative (TN): represent normal/non-malicious instances that were correctly classified as normal/non-malicious.
- False Negative (FN): represent intrusions that were incorrectly classified as normal/non-malicious.

However, confusion matrix are not a very useful tool in comparing different IDSs. To overcome this issue, different performance metrics are defined from the confusion matrix variables, yielding numerical values that can be adopted for comparisons. The defined performance metrics adopted in this study are explained below.

Accuracy is defined as the ratio of all correctly classified instances and the total number of instances:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (4)$$

Precision is defined as the fraction of data instances that are classified as positive and are actually positive:

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

Recall measures the percentage of detected true positives among all the actual true positive traffic:

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

F-score is considered as the harmonic mean of precision and recall as it scores the balance between the two metrics:

$$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (7)$$

In addition to the performance metrics formulated using the confusion matrix variables, the time taken to build and to test each model is also measured. Time to build and test can help identify the better algorithm in terms of rebuilding and analyzing new unseen instances.

B. Experimental Setup

All the experiments are conducted on 1.80GHz Intel (R) Core i7 processor with 8 GB RAM. Python 2.7 and scikit-learn version 0.20.0 [19] installed on Ubuntu 16.04 mate Operating system.

The based machine learning algorithms described in the above subsection are implemented using scikit-learn and implemented for ensemble learning. The ensemble learning methods considered for evaluation are; Adaptive boosting (AdaBoost), Bootstrap aggregation (Bagging), Majority voting (MV), and Random Forest (RF).

V. RESULTS AND DISCUSSION

TABLE II. PERFORMANCE OF ML METHODS IN FLOW-BASED DATASET

No	Algorithm	Performance of ML			
		Accuracy	Precision	recall	f-score
1	DecisionTree	99.09 %	0.99	0.99	0.99
2	NaiveBayes	60.56 %	0.47	0.61	0.48
3	SVC	62.89 %	0.75	0.63	0.49
4	Bagging(DT)	99.08 %	0.99	0.99	0.99
5	Bagging (NB)	60.57 %	0.47	0.61	0.48
6	Bagging (SVC)	62.89 %	0.75	0.63	0.49
7	Adaboost (DT)	99.15 %	0.99	0.99	0.99
8	Adaboost (NB)	70.74 %	0.84	0.71	0.75
9	Adaboost (SVC)	62.35 %	0.39	0.62	0.48
10	RandomForest	99.14 %	0.99	0.99	0.99
11	MajorityVoting	63.44 %	0.76	0.63	0.50

A. Base Classifiers results

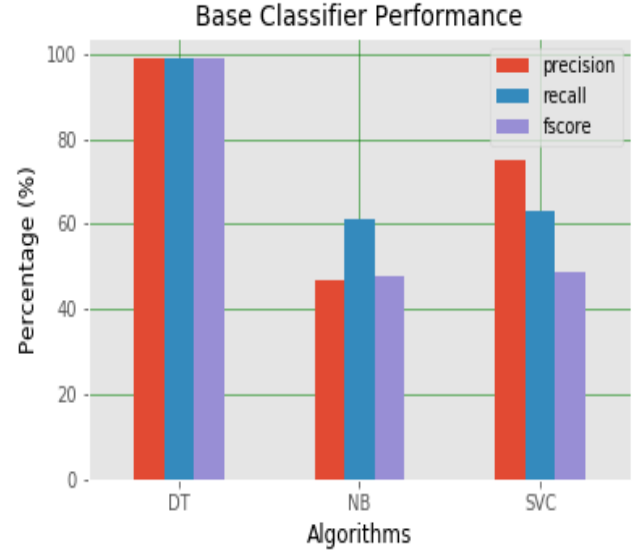


Fig. 5: Base method precision, recall, f-score

Table II present results obtained in this study. The accuracy measure of the based classifiers used in the ensemble methods are decision tree (DT), Naïve Bayes (NB), and Support Vector Classifier (SVC). The better performing base classifier classifiers in detection accuracy is the decision tree (DT) classifier, followed by SVC, then NB at 99.09%, 62.89%, and 60.56 % respectively. The DT base classifier demonstrate outstanding performance, since it is able to classify most of the data correctly.

The precision, recall, and f-score measures of the base classifiers is presented in Figure. 5. The DT acquired precision, recall, and f-score measures above 99 %. On the other hand, NB is the worst performing classifier when compared with SVC base classification method. SVC is able to identify the positive data points better than NB. In terms of correctly identifying the positive points out of the total true positives, SVC and NB almost get values are closer, at 63 % and 61 %. However, the harmonic mean of the precision and recall suggest that between the base classifiers SVC and NB, SVC performs better, see Table II.

B. Bagging Algorithms results

The three base classifiers; DT, NB, and SVC were utilized to construct bootstrap aggregation (Bagging) ensemble learning method. Figure. 6, shows a bar graph representing Bagging (DT), Bagging (NB), and Bagging (SVC). Hence the results are almost identical to the base method results. Bagging (DT) achieve better value, followed Bagging (SVC) which performs slightly better than Bagging (NB).

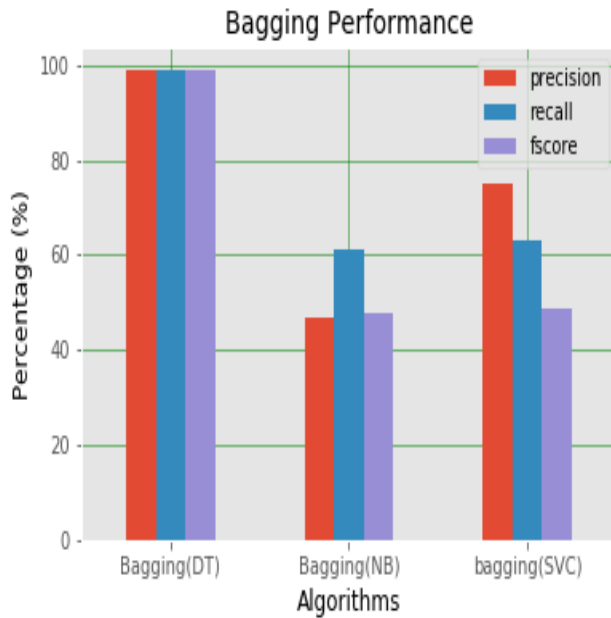


Fig. 6: Bagging precision, recall, f-score

C. Adaboost algorithm results

For the Adaptive boosting algorithm, the results show that Adaboost (DT) achieved better detection accuracy, followed by Adaboost (NB) then Adaboost (SVC), shown in Table II. The Adaboost (DT) also demonstrated better performance in precision, recall, and f-score, see Figure 7.

In conjunction with the Bagging ensemble technique results discussed above, the Adaboost (NB) performs better than Adaboost (SVC). Adaboost (NB) demonstrated better results in terms of correctly identifying all the positive data points out of all the true positives. This method also achieved higher recall and f-score when compared to adaboost (SVC).

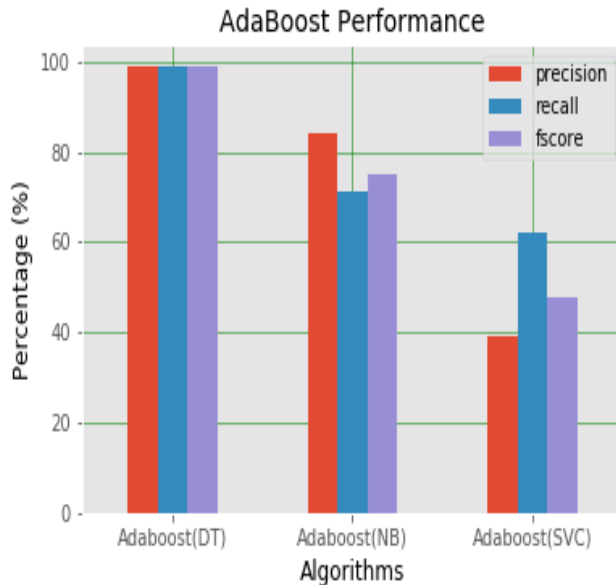


Fig. 7: Adaboost precision, recall, f-score

D. Model Build and Test time

In Figure 8, the classifier build and test times are presented. In terms of build time, the worst performing classifiers are Bagging (SVC), SVC, Bagging (DT), and majority voting. The other algorithms demonstrate a fair training time, hence applicable for IDS scenarios requiring constant system updates. However, when we consider the time each classifier takes to classify a new instant, Adaboost (SVC), Bagging (SVC), and SVC are the worst performing classifiers.

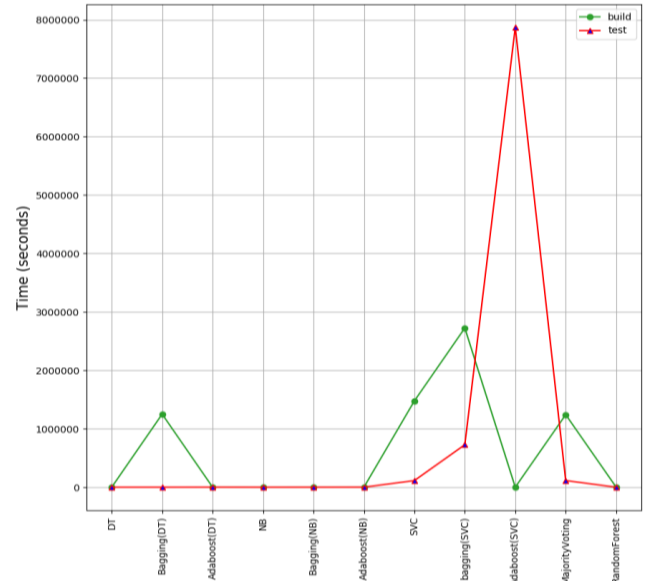


Fig. 8: Classifier build and test time in seconds

E. Discussions

Decision Tree (DT) classification: Methods utilizing the decision tree algorithm tends to have better detection accuracy, and demonstrate better performance in terms of correctly classifying malicious flows.

Probabilistic (Naïve Bayes) classification: Methods utilizing NB demonstrated the worst performance in terms of detecting malicious flows.

Non-probabilistic (SVC) based methods: The results indicate that methods associated with Support Vector Machines perform better than probabilistic approaches in terms of flow classification. In addition, it takes longer to build and to test, hence they are the worst methods to employ when the time is a critical issue.

When we consider the amount of time taken to build the model for each classifier, Bagging (SVC) takes longer to train. This method is followed by the base method SVC, then the Bagging (DT). Such results are evident that SVC based or associated methods take longer to train. The majority voting also took longer. However, this confirms our claim as it contains SVC in its method. The other algorithms require a much more reasonable amount of time to train a model. The results also indicate that Adaboost (SVC) take the longest duration, taking about 7 871 208 seconds when presented with the testing data, as shown in Figure 8. However, the overall results indicated that

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DT based method are the outstanding performers. However, SVC tends to demonstrate better performance when bagging ensemble method is used. On the other hand, NB demonstrated better performance when AdaBoost ensemble method is used. In conclusion, DT based approaches outperform both probabilistic and non-probabilistic classification approaches.

VI. CONCLUSION AND FUTURE WORKS

Network security is a major concern in today's information era. Due to the ever growing need for security, intelligent and efficient IDS are required. Traditional methods of intrusion detection, employ deep packet inspection to analyse data point. This method has limitations and not applicable to high speed networks. This paper proposed a flow based IDS for intrusion detection using ensemble learning methods. To develop a flow-based intrusion detection method using machine learning classifier, an algorithm capable of detection totality and allow to train and retrain in a minimum amount of time is ideal. In this study a performance evaluation of ensemble machine learning methods is conducted. Our approach evaluated Decision tree (DT), probabilistic (NB), and non-probabilistic (SVC) methods employed to bagging and adaboost ensemble methods. The results obtained indicate that DT-based methods can perform better for flow based intrusion detection systems. This is because they perform better in terms of detection accuracy, precision, recall, and f-score. The time taken by decision tree algorithm to build and test a model is reasonably small when compared to the probabilistic and non-probabilistic methods.

A proof-of-concept prototype of a flow based IDS utilising a decision tree based ensemble method (Random Forest) is currently under development. The IDS will be implemented in an SDN wireless network and evaluated in terms of detection accuracy and resource usage. The evaluation will provide more insights into a practical deployment case and performance of the algorithm for intrusion detection tasks using network flow data.

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